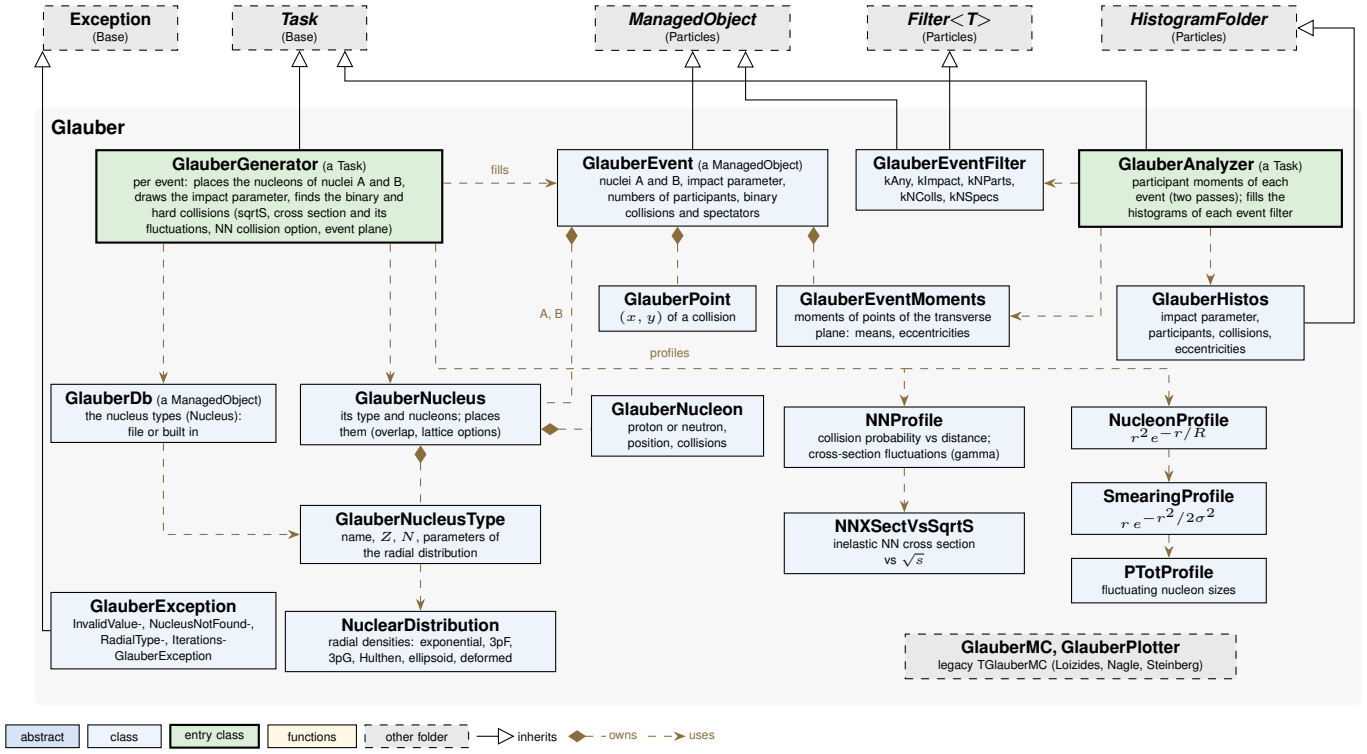

Exception
(Base)
Task
(Base)
ManagedObject
(Particles)
Filter<T>
(Particles)
HistogramFolder
(Particles)
Glauber
GlauberGenerator (a Task)
per event: places the nucleons of nuclei A and B, draws the impact parameter, finds the binary and hard collisions (sqrtS, cross section and its fluctuations, NN collision option, event plane)
fills
GlauberEvent (a ManagedObject)
nuclei A and B, impact parameter, numbers of participants, binary collisions and spectators
GlauberEventFilter
kAny, kImpact, kNParts, kNColls, kNSpecs
GlauberAnalyzer (a Task)
participant moments of each event (two passes); fills the histograms of each event filter
GlauberPoint
(x, y) of a collision
GlauberEventMoments
moments of points of the transverse plane: means, eccentricities
GlauberHistos
impact parameter, participants, collisions, eccentricities
A, B
profiles
GlauberDb (a ManagedObject)
the nucleus types (Nucleus): file or built in
GlauberNucleus
its type and nucleons; places them (overlap, lattice options)
GlauberNucleon
proton or neutron, position, collisions
NNProfile
collision probability vs distance; cross-section fluctuations (gamma)
NucleonProfile
$r^2 e^{-r/R}$
SmearingProfile
$r\, e^{-r^2/2\sigma^2}$
PTotProfile
fluctuating nucleon sizes
GlauberNucleusType
name, Z, N, parameters of the radial distribution
NNXSectVsSqrtS
inelastic NN cross section vs $\sqrt{s}$
GlauberException
InvalidValue-, NucleusNotFound-, RadialType-, Iterations-GlauberException
NuclearDistribution
radial densities: exponential, 3pF, 3pG, Hulthen, ellipsoid, deformed
GlauberMC, GlauberPlotter
legacy TGlauberMC (Loizides, Nagle, Steinberg)
abstract
class
entry class
functions
other folder
inherits
owns
uses